

Friday Session Report

2026-03-13 14:06:57

Total questions: 2

Question: what is a blackhole ?

Answer: A black hole is a region that forms when a mass M is packed into a space whose size (circumference) is smaller than $2\pi r_S$, where $r_S = 2GM/c^2$ is the Schwarzschild radius. In such a configuration an event horizon appears—a boundary from which light cannot escape. (See notes [1] and [2].)

Sources:

- Page 77, Asimov Black Holes - The Collapsing Universe.pdf
- Page 15, Asimov Black Holes - The Collapsing Universe.pdf
- Page 38, Asimov Black Holes - The Collapsing Universe.pdf

Confidence: 84%

Question: how black holes are made ?

Answer: Black holes are created whenever a mass M is packed into a region whose size is smaller than its Schwarzschild radius $r_S = \frac{2GM}{c^2}$. If the circumference of the mass distribution becomes less than $2\pi r_S = 2\pi \left(\frac{2GM}{c^2} \right)$, an event horizon forms and a region from which light cannot escape appears [2]. In practice this can happen in two ways that are mentioned in the notes: **Primordial black holes** – Stephen Hawking suggested that the Big Bang itself could have produced a large number of very small black holes. These would have formed directly from the extreme density fluctuations in the early universe, not from any later stellar process [1]. **Stellar mass collapse** – The classic (Newtonian) picture is that if a star (or any massive object) contracts enough that its escape velocity reaches the speed of light, the object's radius reaches the Schwarzschild value ($r = 2GM/c^2$). At that point light would be trapped and a black hole is formed [3]. (The notes note that no known stellar processes can make the ultra-small black holes discussed in [1], but the general principle of collapse to within the Schwarzschild radius still applies.) Thus, according to the provided notes, black holes are made when enough mass is concentrated inside its own Schwarzschild radius, either in the early universe (primordial black holes) or by the collapse of massive objects such as stars.

Sources:

- Page 92, Asimov Black Holes - The Collapsing Universe.pdf
- Page 77, Asimov Black Holes - The Collapsing Universe.pdf
- Page 23, Asimov Black Holes - The Collapsing Universe.pdf

Confidence: 85%
